Networks and Systems: MM2

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- **Mm1** Introduction to Systems and Networks (hps)
- **Mm2** Layer-2 Example Technologies: Field-buses CAN and Flexray (hps)
- **Mm3** TCP Modeling and Performance Analysis (hps)
- **Mm4** Network Calculus I (HS)
- **Mm5** Network Calculus II (HS)

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Learning goals

- Become acquainted to Field busses
  - typical applications and requirements
- Be familiar with basic properties of CAN and Flexray
  - bus topologies
  - frame layout and payload size
  - Addressing
  - Medium Access
- Understand implications of field bus design decision on communication delays
  - Apply Markov modeling techniques to selected subcases

Field bus application areas

- Industrial communication
  - Production engineering
    - Transmission of programs to computerized numerical control machines
    - Control of plants / automation of car manufacturing
  - Process engineering
    - Control loops in a refinery
    - Control and regulation at aluminum smelting
  - Power generation
    - Conventional thermal power station / nuclear power plant
    - Hydroelectric power plant / pumped-storage power station
- Automotive engineering
    - Distributed real time regulation in cars
    - Commercial vehicles – in-car network
    - Control of special functions in work machines
- Building services engineering
    - Light control in residential houses
    - Air-conditioning technology in functional buildings

fast real-time systems with short deadlines and often high frequencies

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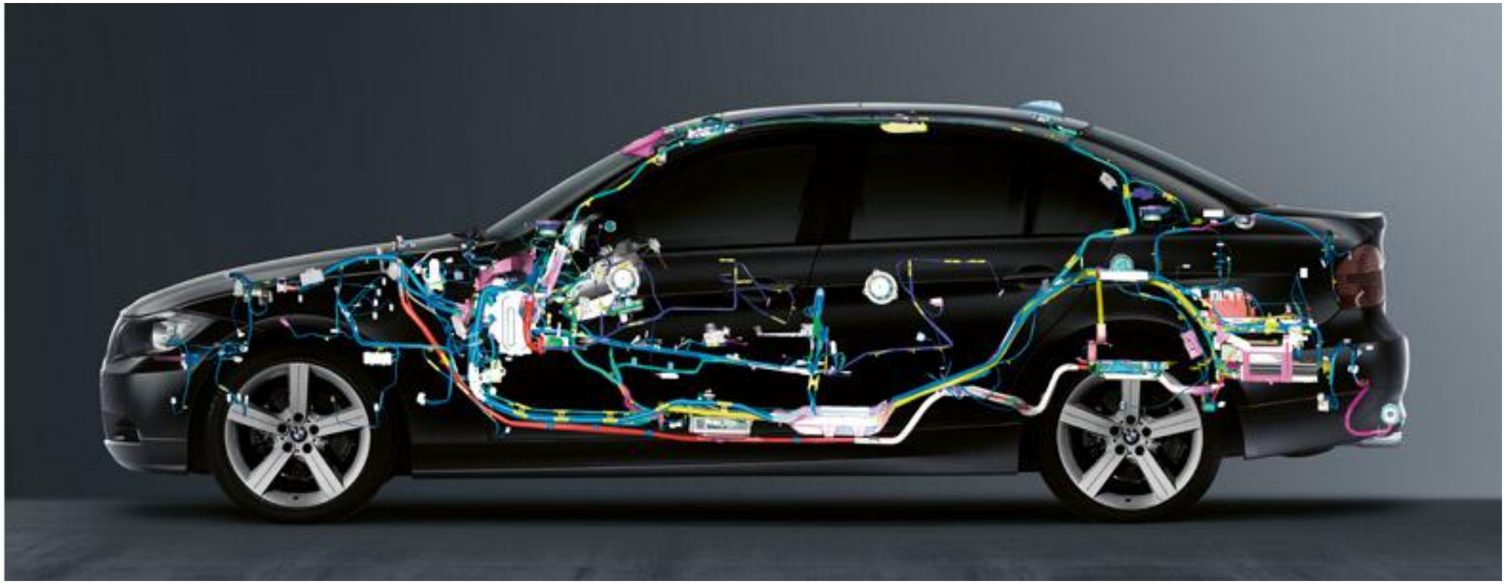
In-car networks

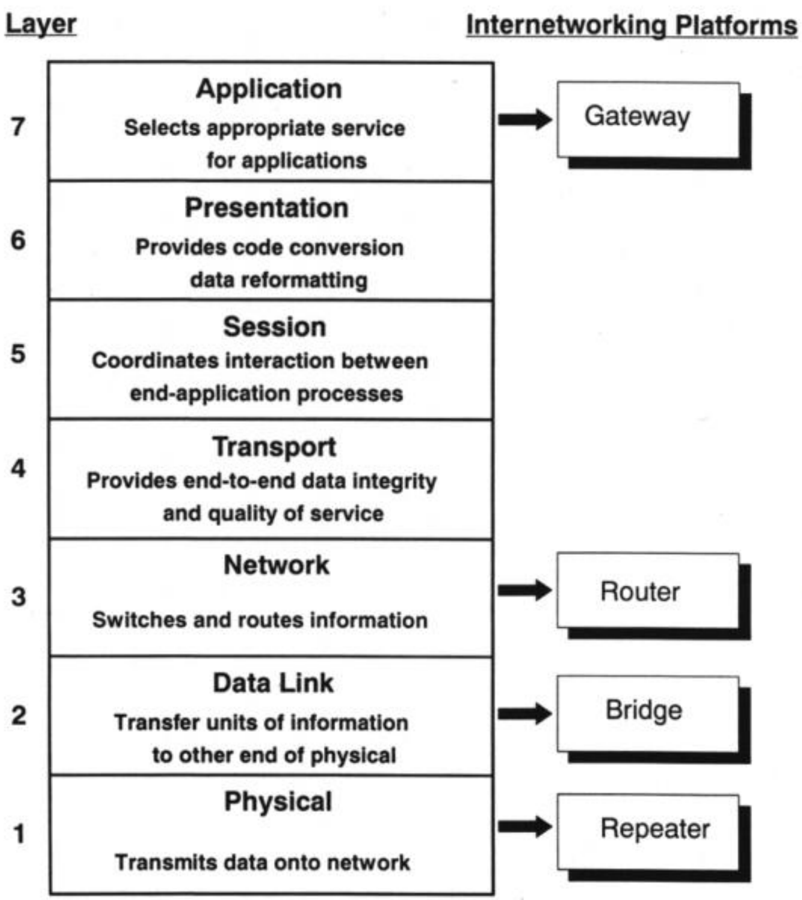


- Today's cars have up to some 100 Electronic Control Units (ECUs) that exchange messages on dedicated communication busses, e.g.:
  - LIN
  - CAN
  - FlexRay
  - Time-triggered Ethernet

In a modern BMW up to 3 km of cable that weigh up to 60 kg.

Source: www.bmw.com (a few years ago)

Fieldbus vs. OSI model



- Fieldbus systems often define several OSI layers in one standard
- Mostly layers 3 to 6 are nonexistent
  - Efficient, fast data processing
  - No routing
  - No fragmentation
- In the majority only layers 1-2 or layers 1-2-7 are defined

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Lecture content

- Intro to field-buses
- CAN
  - Bus architecture
  - Bit transmissions
  - Frame structure
  - Medium Access and Arbitration via dominant ID bits
- Flexray
  - Communication cycle principle: cycle, slot, macrotick, microtick
  - Static and dynamic segments
  - Medium access method in Dynamic Segment
  - Comparison with CAN
- Flexray delay modeling in dynamic segment
  - Using Markov Chain model

Controller Area Network (CAN) at a glance

- Number of nodes
  - unlimited
    (dependent on physical layer)
- Topology
  - line
  - star
- Length of bus lines
  (dependent on transfer rate)
  - 40 m at 1 Mbit/s (specified)
  - 620 m at 100 kbit/s
  - 10 km at 5 kbit/s
- Number of message identifiers
  - 2^{11} (standard frame)
  - 2^{29} (extended frame)
- Data bytes per message
  - 0 … 8
- Bus access
  - CSMA/CD through AMP
    (Carrier Sense Multiple Access/Collision Detection with Arbitration on Message Priority)
  - controlled by message priority
  - non-destructive bit-wise arbitration
- Bus throughput
  - max. 1 Mbit/s (total)
  - max. 577 kbit/s (information/payload)
- Real-time capability
  - guaranteed latency times for high priority messages
    (<134 µs @ 1 Mbit/s)
- Reliability / Safety
  - acknowledgment of message
  - error detection, handling and fault confinement

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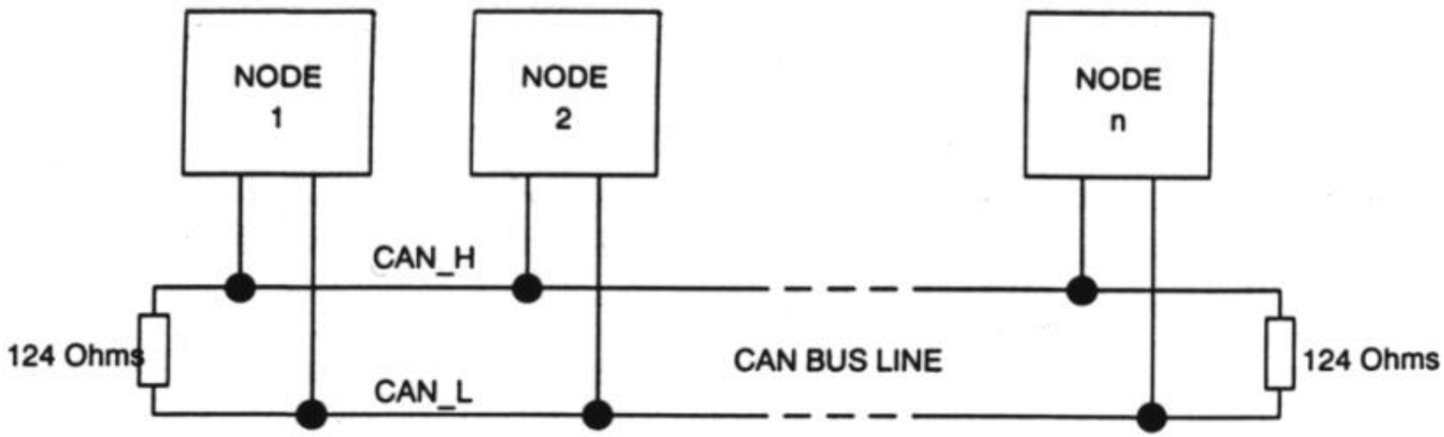
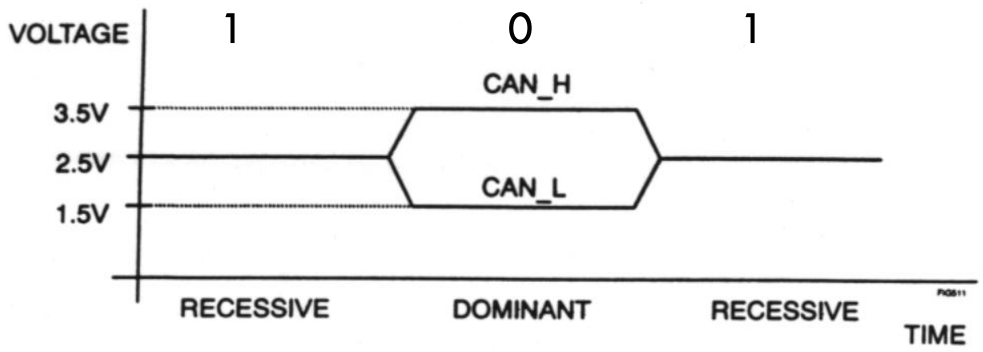
CAN bus architecture and L1





- Bus topology
- Broadcast transmissions
- a bit must "fill" whole bus
  - Limits cable length

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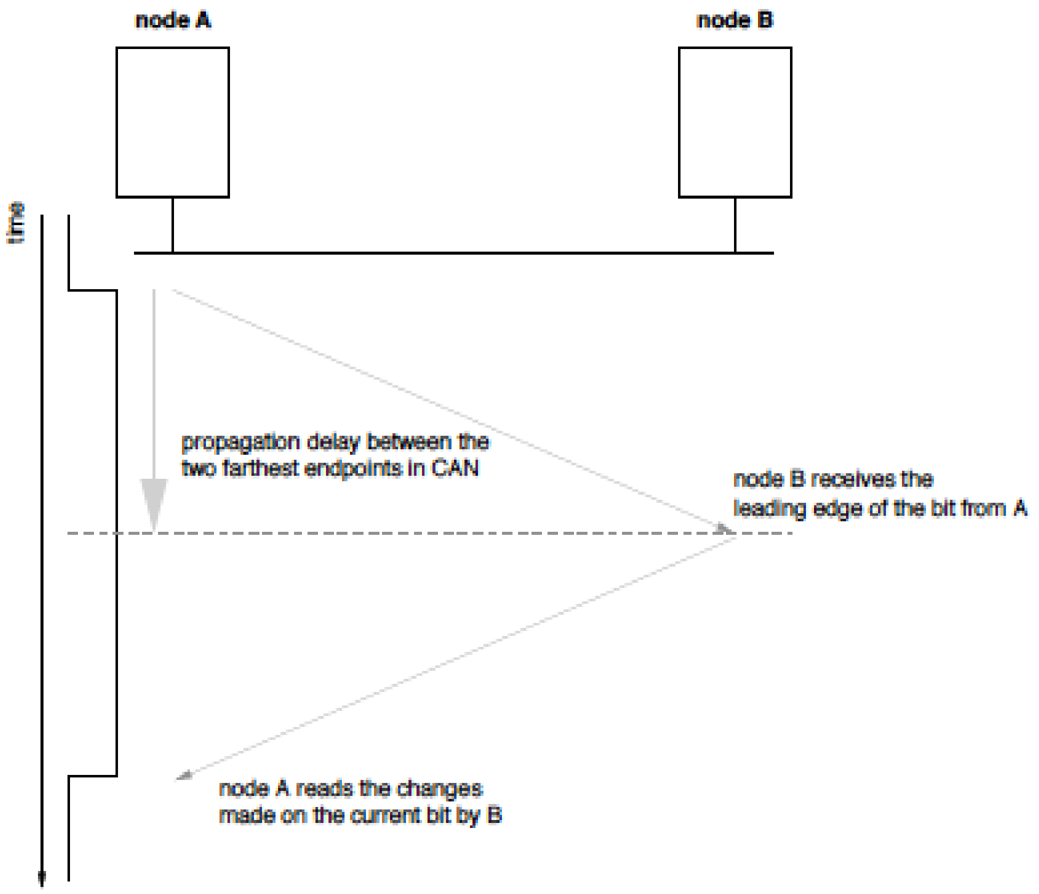
Bit timing

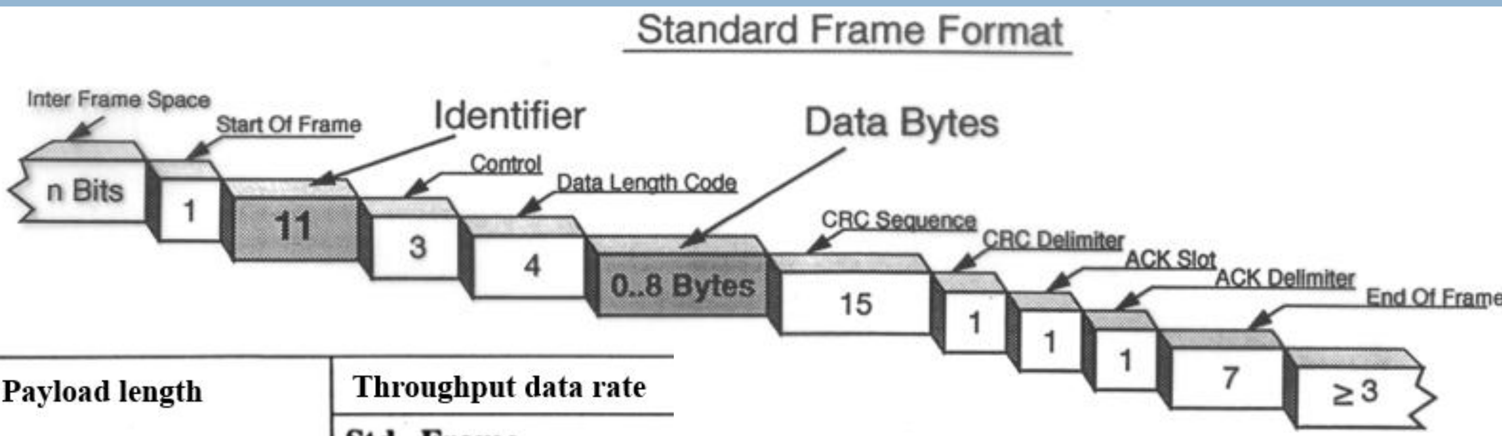
- Nodes need to agree on the value of the bit currently transmitted on the bus
- Synchronization protocol is required (bit-synchronization of nodes)
- Transition edges between the bits are used to resynchronize nodes→ long sequences without bit transitions should be avoided → bit stuffing

Bit timing

- For the purpose of bus arbitration, the nodes should be able to change the status of the transmitted bit from recessive to dominant, and all nodes in the network should be able to detect the change before the bit transmission ends.
- Bit time = at least twice signal propagation time between two the most distant nodes in the network



CAN frame format



Payload length	Throughput data rate
	Std. Frame
0	–
1	72,1 kBit/s
2	144,1 kBit/s
3	216,2 kBit/s
4	288,3 kBit/s
5	360,4 kBit/s
6	432,4 kBit/s
7	504,5 kBit/s
8	576,6 kBit/s

- 11 bit identifier
- 0…8 bytes payload
  - Payload size influences throughput, see left for one example

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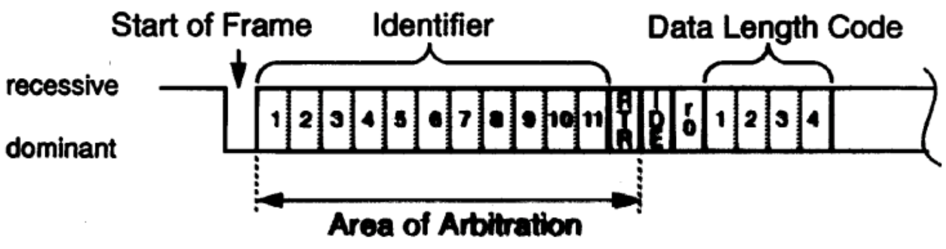
CAN arbitration



TX	Bus (RX)	Action
rec.	rec.	proceed with arbitration
dom.	dom.	proceed with arbitration
rec.	dom.	arbitration lost
dom.	rec.	bit error

- All nodes transmit on shared medium (bus)
- Medium Access scheme: CSMA/CD AMP - Carrier Sense Multiple Access/Collision Detection with Arbitration on Message Priority
  - Start transmission only when medium is idle (else wait for end of previous transmission)
  - Use dominant bits in ID field to avoid collisions
    - Node with larger ID will stop transmission

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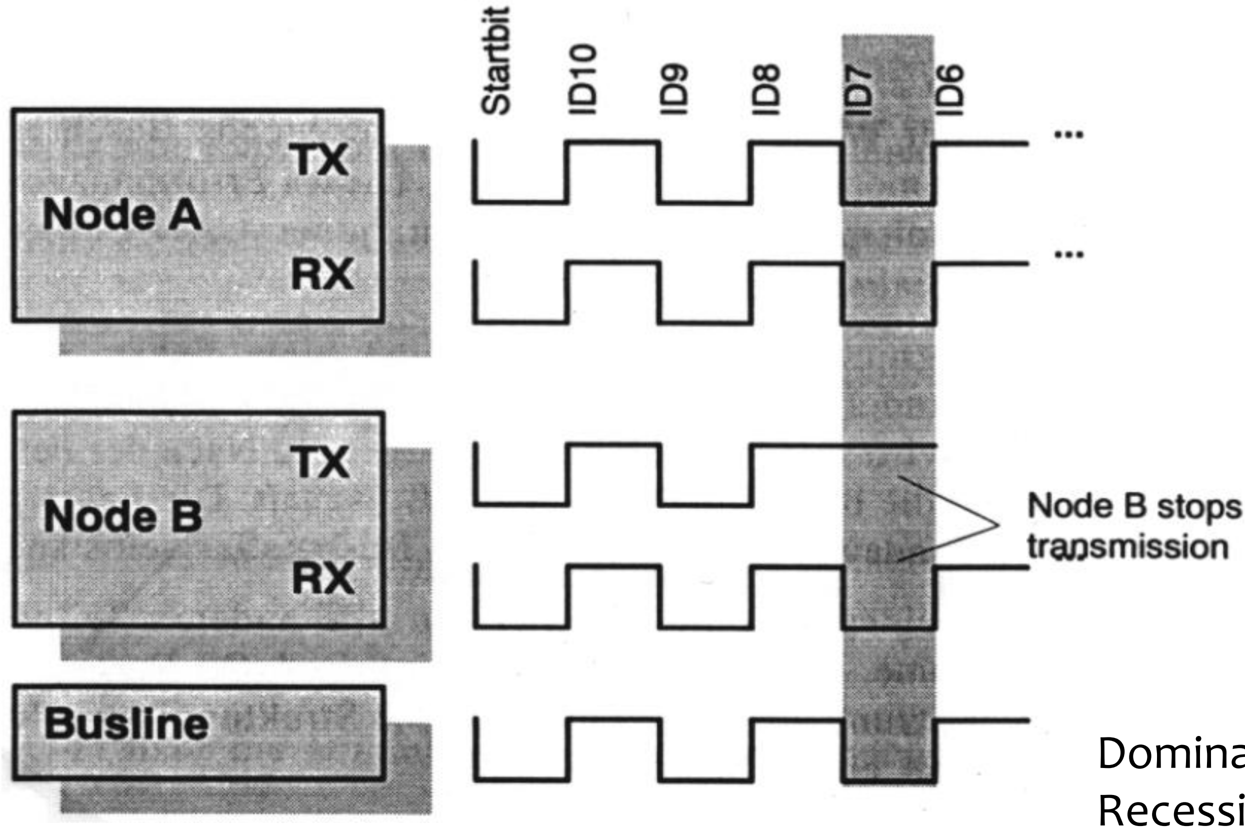
CAN arbitration- example



Dominant bit: voltage on bus
Recessive bit: listen only

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CAN advantages

- Possibility of assigning priority to messages
  - guaranteed maximum latency times for highest priority messages
- Multicast communication with bit-oriented synchronization
- Error detection

Outlook: CAN configuration complexity

Today's CAN networks:

- Hundreds of messages
  - a message is an application input/output, usually 1 CAN ID.
- Multiple and mixed busses (CAN+FlexRay)
  - Connected using gateways (queuing in GWs)
- Latency requirements of down to 5 ms or less (end-to-end!)
- Both periodic and aperiodic traffic (event based)
- Target bus loads of 50% – 60% and even more

- How to ensure requirements?

Verification of CANconfiguration

- Ensure application real-time requirements are satisfied
  - Requirement example:
    - message arrival interval of 5ms ± 10% in 90% of cases.
    - at most 3 consecutive deadline misses.
- Methods:
  - Exhaustive worst-case search
    - Not feasible in complex networks
  - Network calculus (see later sessions)
    - Analytic modeling of traffic flows
    - Takes link capacities and queuing bottlenecks into account
  - Simulation of realistic scenarios
    - Represents typical behavior
    - No guarantees for seeing worst case

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FlexRay overview

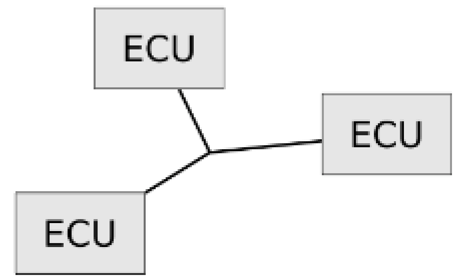
- Fieldbus for automotive applications (cars and other vehicles)
  - Designed to replace CAN for internal communication between ECUs (Electronic Control Units)
    - CAN is not deterministic and has low throughput

- Most important features:
  - High data rates (up to 10 Mbit/s)
  - Time- and event-triggered behavior
  - Redundancy
  - Fault-tolerance (bit-coding, clock-sync)
  - Deterministic

Network Topologies

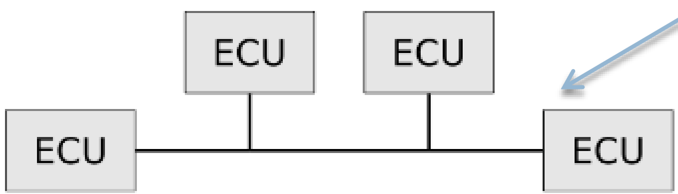
Max point-to-point length is 24 m.

(a) Point-to-point

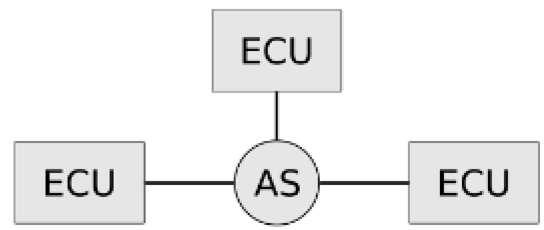


(b) Passive star network

Up to 22 nodes.

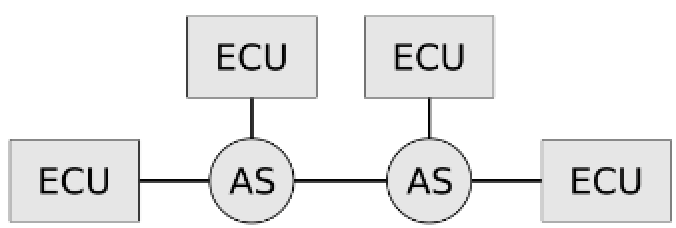


(c) Linear passive bus



(d) Active star network

- For redundancy, each ECU can be connected to two independent channels, possibly with different topologies.

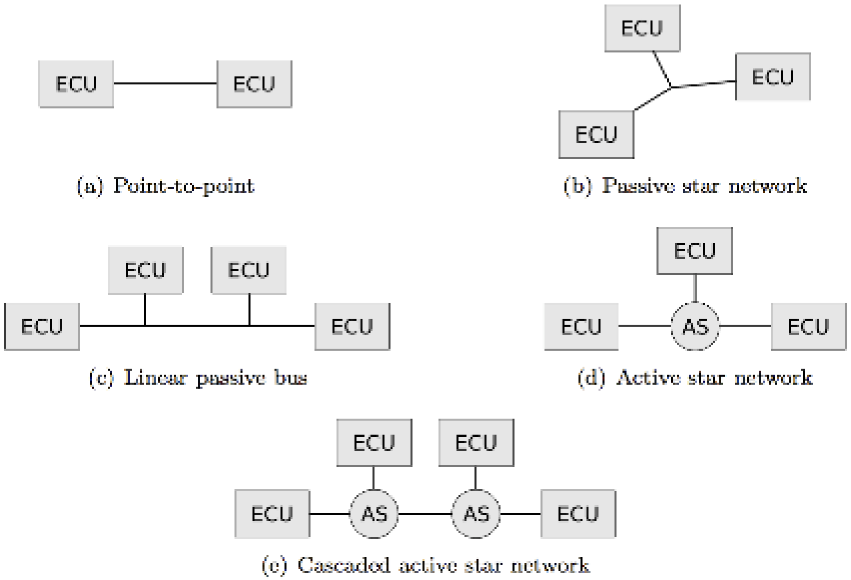


(e) Cascaded active star network

- As many ASs as desired, but adds up to 250ns delay per AS.
- AS replicates all streams to all ports.
- Hybrid topologies are possible.

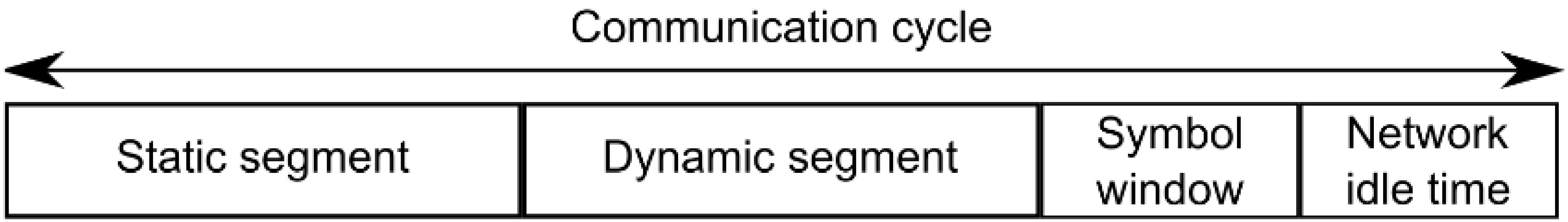
Network Topologies FlexRay vs. CAN

- How is it compared to CAN?
  - Bus segments can be connected with AS. CAN requires GW with re-arbitration on each bus.
  - Bus segment length shorter than CAN (Due to higher bit-rate).



(a) Point-to-point
(b) Passive star network
(c) Linear passive bus
(d) Active star network
(e) Cascaded active star network

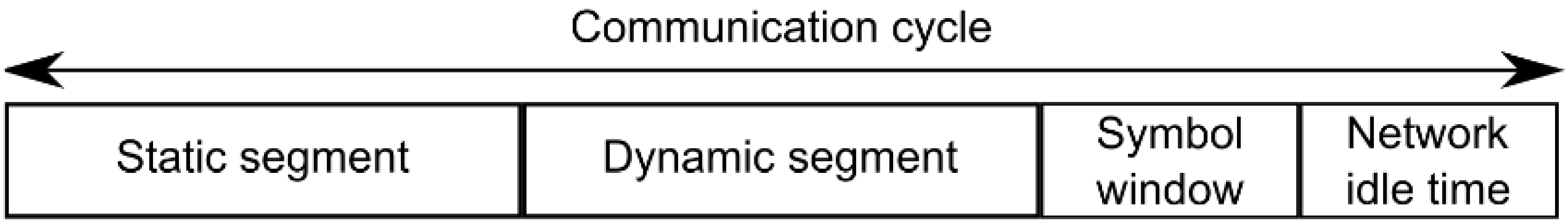
Communication Cycle



- Typical duration of the communication cycle is 2.5, 5, or 10 ms.
  - Determined by shortest periodic requirement in system.
- **Static segment** is transmitted first.
  - Each static frame is transmitted in one static slot in the static segment.
  - The duration of a slot in the static segment is the same for all slots (gdStaticSlot).

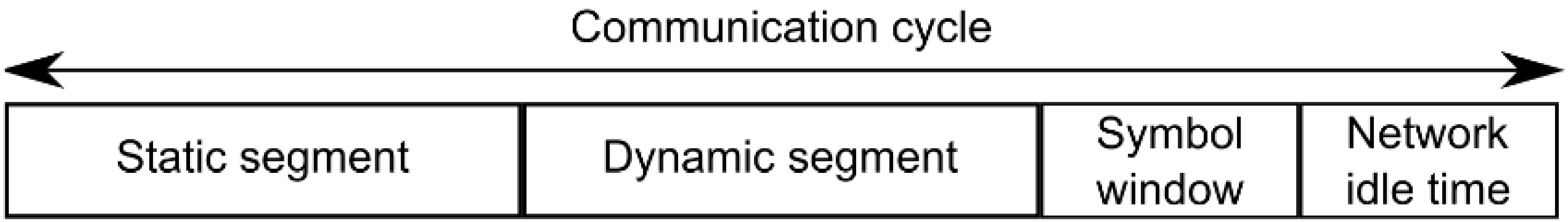
- An example of a static application could be a distributed control application like Anti-lock Braking System (ABS) where the tire position is sampled continuously.
- When the pre-defined duration of the segment has elapsed, the frames in the dynamic segment are transmitted.

Communication Cycle



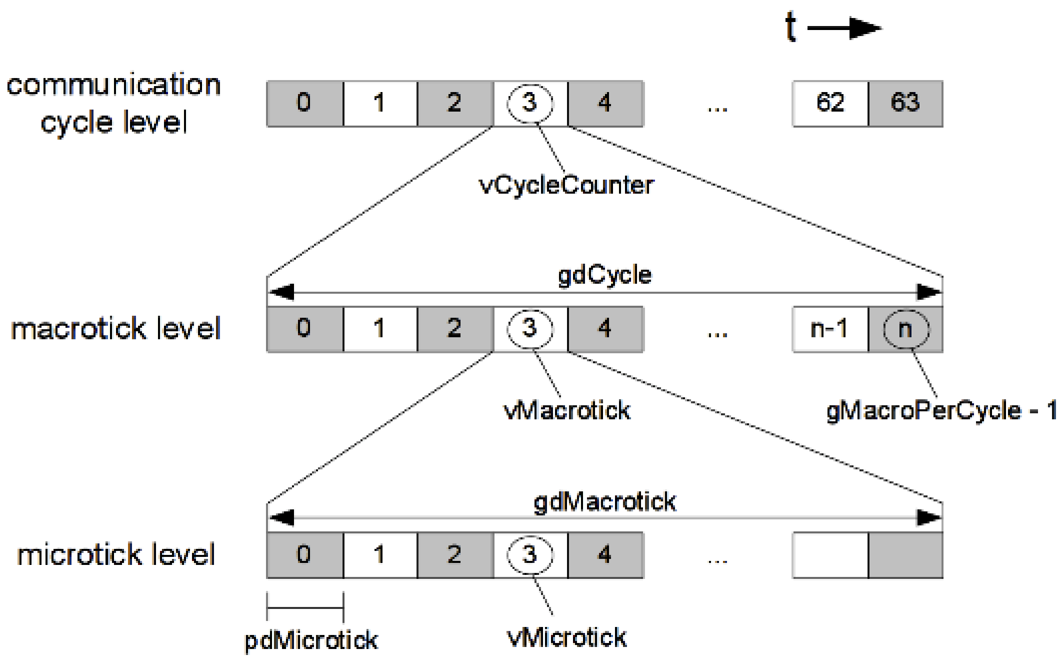
- Static segment
- **Dynamic segment**
  - The dynamic segment uses a dynamic minislotting scheme for medium access.
  - Prioritized access until segment ends.
  - Variable frame sizes.
- Is typically used for event based applications like for example the turn signal on a car that is only activated on user request.
- **Symbol window:** Special bit patterns, e.g., for waking up nodes.
- **Network idle time:** Unused time where time sync takes place.

Communication Cycle FlexRay vs. CAN



- How is it compared to CAN?
  - CAN has no cycle – it is purely event-based.
  - Dynamic segment is somewhat similar to CAN (prioritized access)

Timing Hierarchy



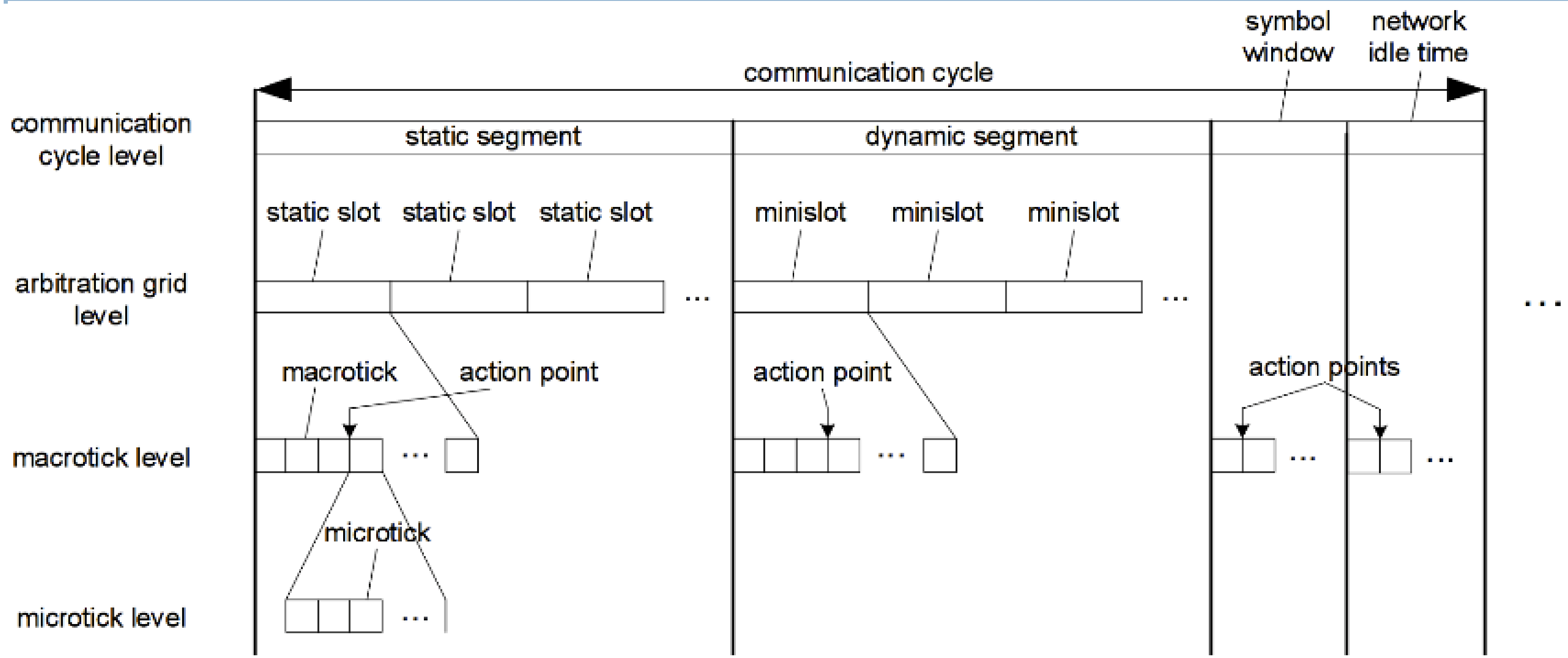
- Same number of macroticks per cycle:
  - in all cycles, and
  - in all nodes.
- At any given time all nodes should have the same cycle number.

- Constant no. of macroticks in cluster and cycles.
- Nodes should have same count.
- Duration can be non-integral number of microticks.

- Controller specific.
- Duration may vary between controllers.
- Internal time granularity is microtick.

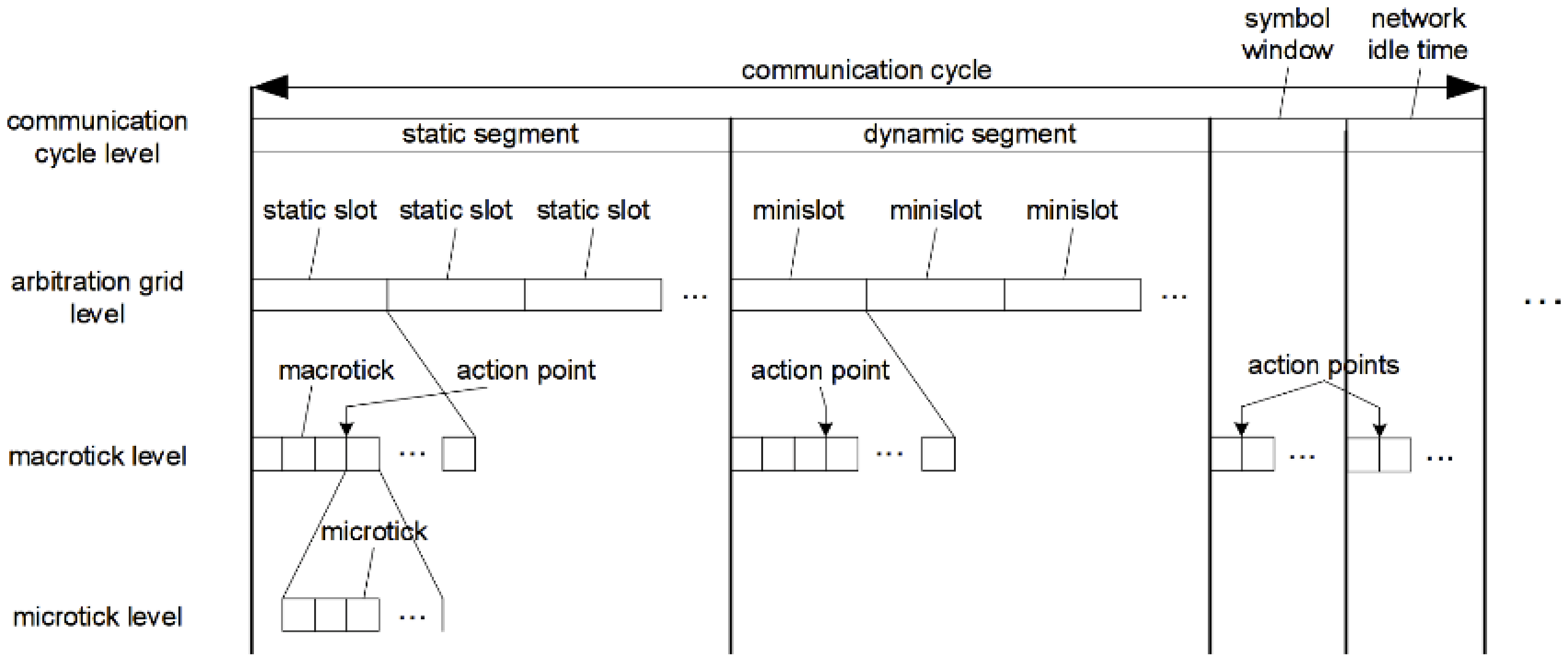
Cluster-wide definitions		
gdCycle	Length of a communication cycle.	$10 - 16000\mu s$
gMacroPerCycle	Number of macroticks in a cycle.	10 – 16000 MT
gdMacrotick	Duration of a macrotick.	$1 - 6\mu s$
pdMicrotick	Duration of a microtick.	$0.0125 \ldots 0.05\mu s$
gdBit	Nominal bit time.	$0.1 \ldots 0.4\mu s$
Variables in each FlexRay node.		
vCycleCounter	Current value of cycle counter.	$0 - 63$
vMacrotick	Current value of the macrotick counter.	$0 \ldots gMacroPerCycle - 1$
vMicrotick	Current value of the microtick counter.	-

Communication Cycle



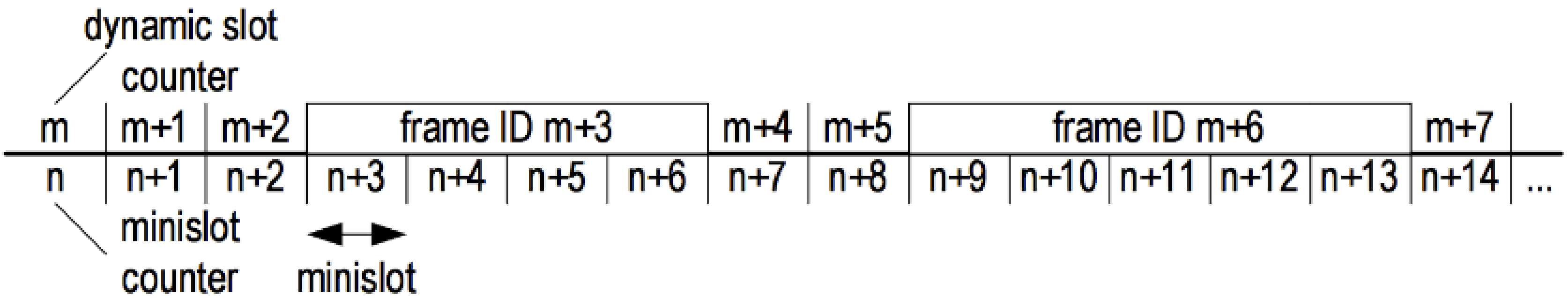
- **Static segment**
  - Each static frame is transmitted in one static slot in the static segment.
  - The duration of a slot in the static segment is the same for all slots (gdStaticSlot).
  - The transmission of static slots shall start at an action point instant (boundary of a macrotick).
  - The static segment uses a static Time Division Multiple Access (TDMA) medium access scheme, which means that an application must be pre-assigned a static slot in the communication cycle.

Communication Cycle



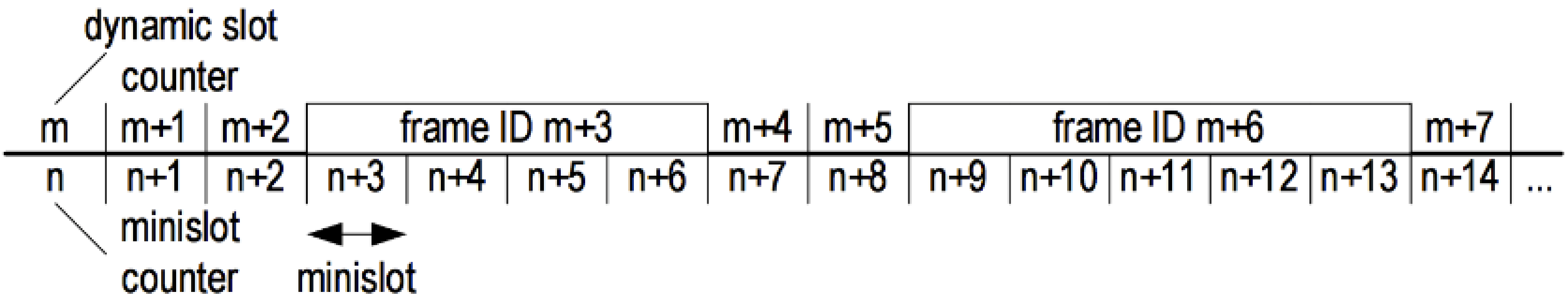
- **Dynamic segment**
  - The transmission of a dynamic frame takes place in a dynamic slot that takes up an integer number of minislots in the dynamic segment.
  - Variable frame sizes.
  - The dynamic segment uses a dynamic minislotting scheme for medium access.
    - Prioritized access until segment ends →Not all are guaranteed to transmit.

Dynamic minislotting



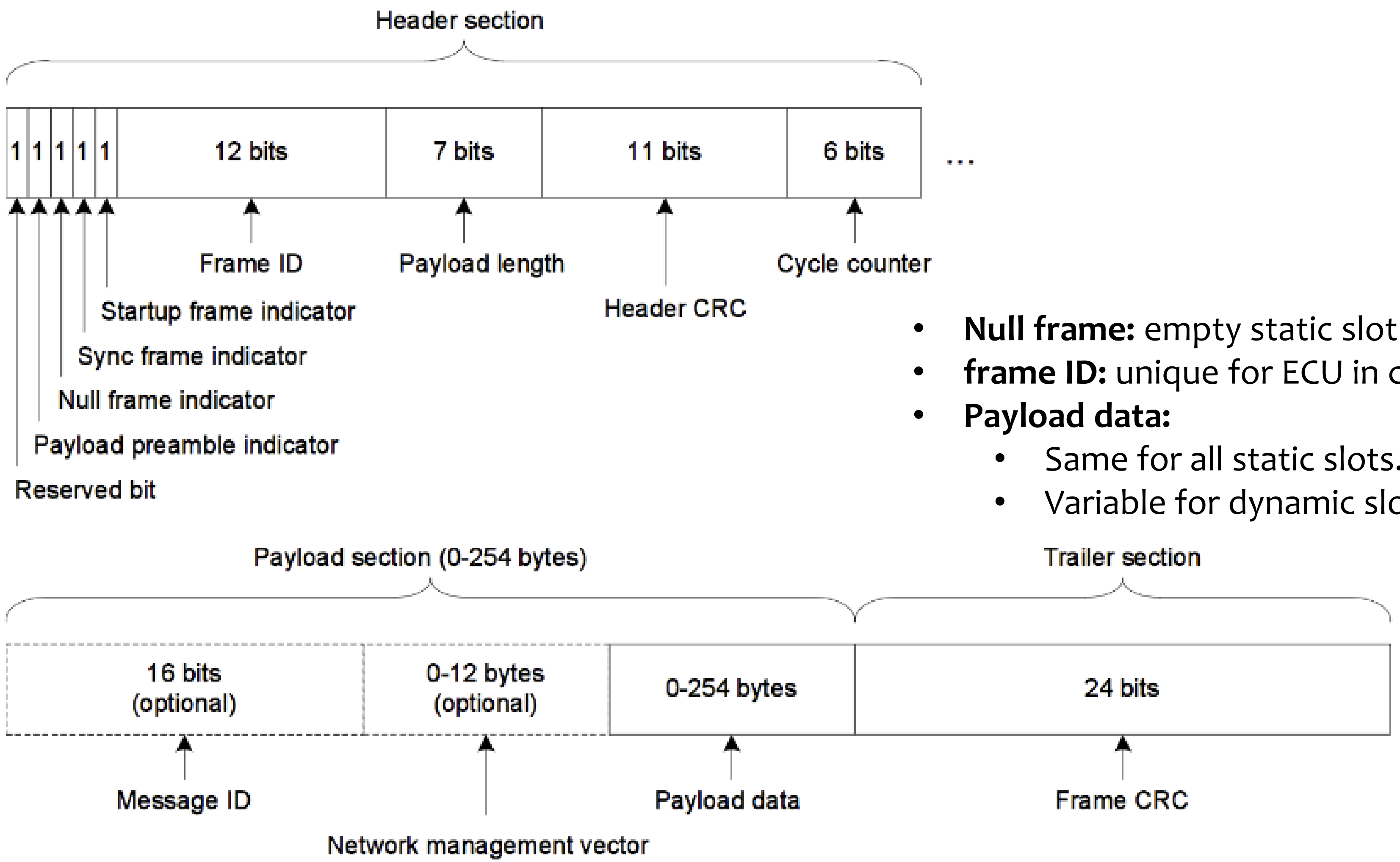
- All nodes have unique priorities = frame IDs when transmitting.
- Each node keeps track of the current slot number. The node is allowed to transmit a frame when the value of the slot counter matches the frame ID, illustrated by m + 3 and m + 6.
  - Dynamic slot counter is constant while transmitting (over multiple minislots).
  - If no frame is transmitted, the dynamic slot only last for a single minislot.
- The number of minislots in each cycle gNumberOfMinislots is fixed.
- Depending on how many frames are being transmitted in a cycle, the slot counter will reach different values after the gNumberOfMinislots available minislots have been used.
  - The more/longer frames, the lower the end value of the slot counter.
  - Each node has a pre-defined parameter pLatestTx that contains the slot ID of the last minislot in which a whole frame can be successfully transmitted by that node. When the slot counter exceeds the value of pLatestTx, the node will not be able to transmit in the current cycle. Any pending frames in the transmit buffer will thus be displaced to the next cycle.

Dynamic Minislotting FlexRay vs. CAN



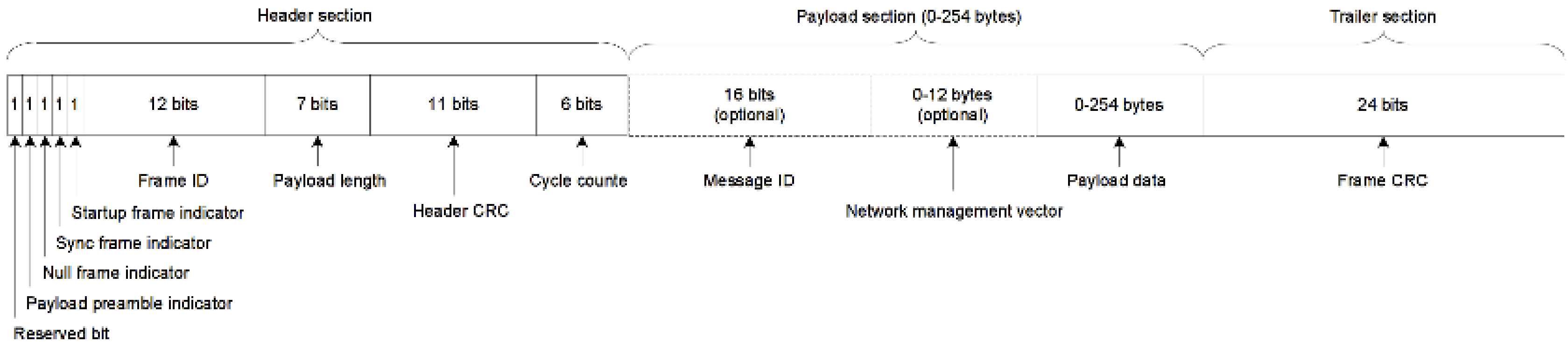
- How is it compared to CAN?
  - Similar: prioritized access, lowest priority can be forced to wait.
  - Difference: no transmission still takes 1 minislot

Frame Layout



- **Null frame:** empty static slot
- **frame ID:** unique for ECU in cycle.
- **Payload data:**
  - Same for all static slots.
  - Variable for dynamic slots.

Frame Layout
FlexRay vs. CAN and Ethernet



- How is it compared to CAN?
  - 12 bits frame ID vs. 11 bits ID in CAN
  - Frame ID relates to message and is used as priority in dynamic segment.
  - CAN only 0-8 bytes, but payload size can always vary. FlexRay payload size is only variable in dynamic segment.
- How is it compared to Ethernet?
  - Ethernet is 0-1500 bytes, variable.
  - Dest. address instead of frame ID.

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Performance aspects

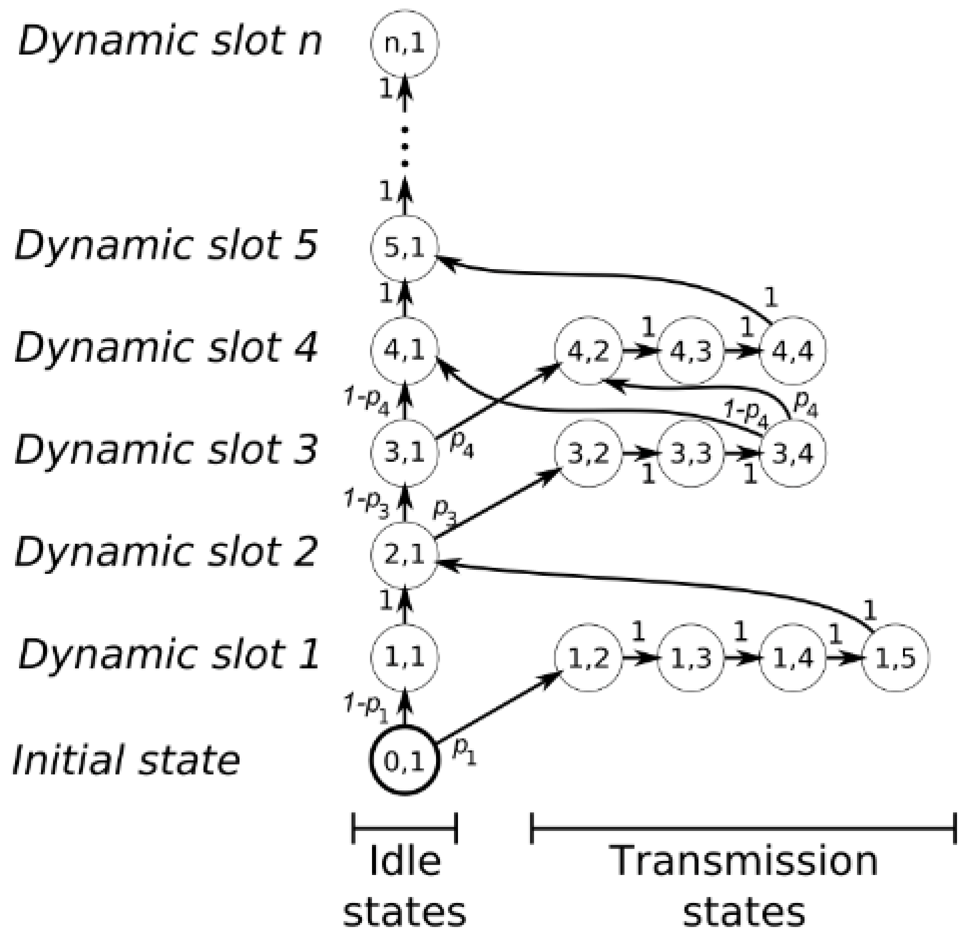
- Throughput
  - For sensor access, throughput is often not first design priority
  - In in-car networks, for e.g. software update in the nodes, high throughput is desired.
  - In the static segment, throughput of a slot is expressed by the cycle time and payload length.
    - Since all slots are same length, a long payload, gives similar increase in the cycle time. (for fixed number of nodes)
    - How to get higher throughput for an ECU?
      - Multiple slots assigned can give high throughput.
  - In the dynamic segment, a high throughput in a slot can be achieved by using a long payload length.
  - But, long frames lead to displacement of later frames, so long frames should probably come last.
- Delay/jitter
- Redundancy
- Loss

Performance aspects

- Throughput
- Delay/jitter
  - Static segment:
    - completely deterministic; and delay maximally until next cycle
  - Dynamic segment:
    - **Displacements to later cycle can occur → Delay**
    - Dynamic slot placement changes → possible larger Jitter
- Redundancy
  - Redundancy in space by using a 2-channel configuration
    - minimizes the risk of errors caused by electrical interference or cable fracture.
  - Redundancy in time can be achieved by transmitting more often than actually required,
    - e.g. in every cycle or by assigning multiple slots to the same application.
- Loss
  - No automatic re-transmissions in FlexRay: the individual applications should handle loss in the most suitable way.

Dynamic Segment State Machine

- A cycle in the dynamic segment can be modeled as a state machine, with parameters:
  - n minislots
  - frame length
  - transmission probability
- Count number of steps, stop when it exceeds *gNumberOfMinislots*
- State model can be used in Discrete-Time Markov Chain, see paper:
  - Jimmy J. Nielsen, Amen Hamdan, Hans-Peter Schwefel, *Markov chain-based performance evaluation of FlexRay dynamic segment*, 6th Intl WORKSHOP On Real Time Networks (RTN 07)



Note: In the shown example, Frame ID 2 and 5 and higher never have any data. Variants of state diagrams without explicite idle states are also possible

Discrete Time Markov Chain Model

- Assumptions
  - Constant frame length for each frame ID
  - i.i.d. frame arrival probability with value p_i for frame_id i
- Obtain displacement probability
  - *gNumberOfMinislots* minislots in the dynamic segment
  - Calculate state probability of discrete Markov chain after k= *gNumberOfMinislots* steps: $\pi(k) = \pi(0)\mathbf{P}^k$
  - Displacement prob: $$P(s \text{ is displ.}) = \sum_{b=2}^{l_s} \pi(k)_{s,b} + p_s \cdot \sum_{a=1}^{s-1} \left(\sum_{b=1}^{l_{max}+1} \pi(k)_{a,b} \right) \quad (3)$$

Summary

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Exercises

1. Discuss advantages/drawbacks of using CAN as opposed to Flexray, when assigning messages in static segment vs. dynamic segment for the latter. What technology and configuration would you use for:
   - periodic/event based messages
   - high/low delay or jitter requirements
   - high/low throughput

2. Draw a dynamic segment scheduling diagram for the three cases below and determine which of the following messages are transmitted:
   Frame lengths are A=5, B=7, C=3, D=4 minislots.
   Assume that there are 40 minislots available in the dynamic segment.
   - **Case 1:** Frame IDs are: A:1, B:2, C:3, D:4
   - **Case 2:** Frame IDs are: A:1, B: 11, C:21, D:31.
   - **Case 3:** Same as case 2, but D:25.

3. Let's assume there are 10 minislots in the dynamic segment. Frame_IDs 1, 3, 5, 7, have frame lengths 3,2,4,3 and data is there with probabilities 0.3,0.2,0.4, 0.3, respectively. Set up the Markov chain model for this case, determine the P matrix and calculate the displacement probability for frame ID 7.